

zontally into “peripheral” things (think actual computer peripherals, such as printers, or spin-offs such as mobile phones). This is nothing but the edifice—not monolithic, not entirely networked, not entirely fuzzy either, but just one complex ecosystem—called information technology.

- (b) The possibility of more and more advanced data manipulation leads to more and more possibilities in the solving of human needs.
- (c) Advanced manipulation also leads to the creation of new needs—that’s why tech advances.

And there we stand today, collectively supported by this framework of what may loosely be termed “networked information”; and that’s what all that follows is about—how we came to stand on and depend upon this framework.

1.2 Before Babbage

Circa 500 B.C.

The abacus is probably what most people associate with the phrase “ancient counting machine.” It’s not entirely ancient—schoolchildren in countries such as China and Japan are still taught how to use it for basic arithmetic operations. Interestingly, skilled abacus users can even find square and cube roots of numbers using the abacus!

You probably know what an abacus looks like—it’s a frame with vertical rods, a horizontal crossbar, and beads. It is based on the decimal system: the first vertical bar is for units, the second for tens, and so on. Space does not permit a discussion of how the abacus works, and you’d do well to visit <http://science.jrank.org/pages/2/Abacus.html> for an explanation.

The device was first used in Babylon around 500 B.C., but the form of the abacus most people (who have seen it!) are familiar with was first used in China around the year 1300.